

Compressing Global Dynamics of Deep Nonlinear Feature Learning

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Abstract

How much can the training dynamics of a deep network be compressed? A snapshot of a trained network is easy to compress; its training is not, because networks that agree on every input can drift apart once training resumes. We ask how small a state suffices to reproduce the *whole* training trajectory while keeping depth, effective nonlinearity and feature learning, because the state that survives is a candidate for the variables that govern learning. Existing exact descriptions of feature learning compress *width* but not *time*: Tensor Programs and dynamical mean-field theory keep the full history of forward and backward responses, and we collect evidence that exact capture is possible only through operator equations that carry the whole learned matrix. We show that the history itself is compressible. Measured in a clock that runs at the speed of learning, each neuron’s forward and backward response history is summarized by P shifted-Legendre moments. The resulting autonomous system has a state that does not grow with training time and, for networks of any depth, tracks the whole parameter trajectory with error $O_T(P^{-1})$, or $O_T(P^{-2})$ with a second, joint clock. The mechanism is that pairing forward and backward memories makes the error in each weight update a product of two projection errors. As a corollary, on any finite horizon every learned weight increment lies within $O(1/P)$ of a rank- mP matrix built from the network’s own response moments. On circle and MNIST tasks, orders $P = 2$ – 3 reproduce the dense predictor over the whole input space, while frozen or directly trained low-rank bases of the same size do not.

1 Introduction

A missing model of learning. The Turing machine is the canonical model of computation. It is idealized—it has no limit on memory or precision—and it is simple enough to reason about, while physical computers, from the von Neumann architecture onward, are understood as its finite-resource implementations. One may argue that the Turing machine is a model of computing rather than of learning, and that learning still lacks its canonical idealized model. Deep neural networks trained by gradient descent are the obvious candidates for implementations of such a model, but the object they would implement is not yet known. The obstacle is that their training dynamics are enormously complex and have no simple representation.

Why look in the limit. A natural place to look for the idealized object is the limit of infinite width. Under feature-learning scalings of the type used in tensor-program and mean-field theory (Yang and Hu, 2021; Nguyen and Pham, 2023), width and learning rate are absorbed into the parametrization and the quantities we care about—predictions, kernels, the laws of hidden neurons—evolve deterministically. The limit is the network with no resource constraint, as the Turing machine is the computer with none. A finite network would then be understood as this limit plus a quantified deviation from it, a CLT-style picture in which each finite network combines its idealized limit with fluctuations around it.

Why compression. Going to infinite width to simplify seems paradoxical: the dynamical system grows with width and becomes infinite in the limit. Statistical physics resolves the paradox in other settings. A gas of 10^{23} molecules is simpler to describe than a gas of ten, because its macroscopic behaviour is captured by a few collective variables; mean-field theory replaces intricate particle interactions by smooth population fields. For training, however, the limit is infinite in two directions: in the number of neurons, and in time, because the state of training depends on the whole history of how each neuron has responded. Mean-field and tensor-program limits tame the first. Whether the second can be tamed as well is the question of whether the limiting training dynamics can be *compressed*. We do not seek compression for its own sake. A compressed description must retain what drives learning and may discard the rest, so the state that survives is a candidate for the variables that govern learning, and a compressed system that reproduces training would be a step toward the idealized model.

Training, not snapshots. Compressing a snapshot is easy: any representation of the current input–output map will do. But two networks that compute the same function on every input can have different Jacobians and backward responses, and once training resumes the discrepancy accumulates. An idealized model of learning must reproduce the whole trajectory.

What the model must keep. Several idealized models of training already exist, and each gives up one of three properties. Deep linear networks admit exact solutions but lose the nonlinearity (Saxe et al., 2014); mean-field analyses are largely confined to one hidden layer (Chizat and Bach, 2018; Mei et al., 2018; Sirignano and Spiliopoulos, 2019); and the neural tangent kernel keeps depth and nonlinearity but freezes the features (Jacot et al., 2018; Lee et al., 2019; Chizat et al., 2019). Each is simple precisely because of what it gives up. We keep all three: *depth* (more than one hidden layer), *nonlinearity* that stays effective during training and in the limit, and *feature learning* (hidden representations move at leading order). Together with the population limit, this contract preserves what is hardest about deep learning while making it a single well-defined object.

The gap. Existing exact descriptions of this object do not compress it. Tensor Programs carry variables indexed by every earlier training step (Yang and Hu, 2021), and feature-learning dynamical mean-field theory (DMFT) evolves two-time kernels whose memory grows quadratically with the number of time points (Bordelon and Pehlevan, 2022). Both are remarkable statements about the network’s behaviour, and both are exact; but they avoid storing the learned matrix only by storing the history of forward and backward responses from which it is built. The information is moved rather than compressed. In the terms above, these descriptions tame the infinity in *width*—their state does not depend on the number of neurons—but not the one in *time*: their state grows with the length of training. No result we know of shows that deep nonlinear feature learning can be captured by a system whose size stays constant throughout training.

Results. Section 4 makes compression of dynamics precise and distinguishes three classes of state: scalars, populations and operators. We then show the following.

1. *Exact capture points to operators* (Section 5). Three pieces of evidence—an exact operator closure for deep linear networks, an exact but never-closing hierarchy of current observations for a two-hidden-layer network, and a restricted failure of scalar closures—suggest that exact capture of deep nonlinear feature learning requires operator equations, which compress nothing. We do not prove this.

2. *Training history is compressible in time* (Section 6). For networks of any depth, replacing each neuron’s forward and backward history by P Legendre moments in a learning-speed clock gives an autonomous system whose state does not grow with training time and whose trajectory is within $O_T(P^{-1})$ of gradient flow, or $O_T(P^{-2})$ with a joint clock (Theorems 6.1 and 6.2). The difficulty is feedback: the memorized signals depend on the approximation itself.
3. *A structural consequence* (Corollary 6.3). Every learned weight increment is, on a finite horizon, within $O(1/P)$ of a rank- mP matrix built from the network’s own response moments.
4. *Evolving memories are what matter* (Section 7). Small orders reproduce the dense predictor over the whole input space, while, in our comparisons, frozen dictionaries and directly trained low-rank corrections of the same size fail.

The theorems hold at every fixed width. The population limit is the motivation and gives an interpretation of the result (Section 6.6), and a precise conjecture (Conjecture 6.4) states what a population version would be; it is not proved here. Appendix C records an earlier, weaker closure of the population flow itself, in which the learned middle layer is confined to a frozen span of initialization observables; it converges without a rate, and its limitations motivated the evolving memories of the main result.

2 Related work

We organize prior work by the idealized picture of training it offers, by what that picture gives up, and by what it compresses: width, time, or neither.

Idealizations that drop one property. Exact solutions of deep linear networks keep depth and feature learning but not nonlinearity (Saxe et al., 2014). Wasserstein and mean-field limits describe one-hidden-layer networks as the gradient flow of a measure over neurons (Chizat and Bach, 2018; Mei et al., 2018; Sirignano and Spiliopoulos, 2019); they keep nonlinear feature learning but not depth. The neural tangent kernel and lazy training keep depth and nonlinearity but freeze the features, reducing training to a fixed kernel (Jacot et al., 2018; Lee et al., 2019; Chizat et al., 2019). Each is simple because of what it sacrifices; our contract rules out all three simplifications.

Exact limits that keep everything. Tensor Programs IV proves feature-learning limits at each fixed number of steps, with a recursion that includes variables indexed by every earlier step (Yang and Hu, 2021). Feature-learning DMFT evolves two-time activation, gradient and response kernels, with memory quadratic and time cubic in the number of time points (Bordelon and Pehlevan, 2022), and its finite-width extension adds perturbative fluctuations (Bordelon and Pehlevan, 2023). Multilayer mean-field theory and Neural Feature Flow evolve fields on neuronal ensembles (Nguyen and Pham, 2023; Fang et al., 2020). These keep all three properties and identify the idealized object. They compress width: DMFT, for instance, has a state independent of the number of neurons. But their state grows with the length of training, or consists of infinite-dimensional fields. Our result is complementary: it keeps the neurons and compresses time.

Finite hierarchies. The Neural Tangent Hierarchy derives an exact hierarchy of outputs and higher tangent kernels and truncates it at order p (Huang and Yau, 2020). An order- r kernel on m training points has up to m^r entries, so storage through order p grows like m^p (a direct count rather than a statement of that work), and its guarantee holds on a width-dependent horizon with error

decreasing in width rather than in the order. Response memory instead controls the full parameter trajectory at fixed width, with a state linear in P and error decreasing polynomially in P .

Memory and low rank. HiPPO provides online Legendre projection of a given signal (Gu et al., 2020); we apply it to endogenous, coupled training histories and prove stability of the resulting feedback loop. Low-rank training methods constrain the learned increment to evolving factors (Zhao et al., 2023); response-memory factors are causal summaries of the actual forward and backward histories, and a directly trained factorization of matched rank performs far worse (Section 7).

3 Setting

Fix m training pairs (x_a, y_a) with $x_a \in \mathbb{R}^d$, width n and $H \geq 2$ hidden layers:

$$\begin{aligned} z_a^{(1)} &= W^{(1)} x_a / \sqrt{d}, & h_a^{(1)} &= \phi(z_a^{(1)}), \\ z_a^{(\ell)} &= W^{(\ell)} h_a^{(\ell-1)}, & h_a^{(\ell)} &= \phi(z_a^{(\ell)}), \quad 2 \leq \ell \leq H, \\ f_a &= \frac{w^\top h_a^{(H)}}{n}, & r_a &= f_a - y_a, \quad \rho = \left(\frac{1}{m} \sum_a r_a^2 \right)^{1/2}. \end{aligned} \tag{1}$$

The activation ϕ may differ between layers; the loss is $\mathcal{L} = \rho^2$. The backward responses, which exclude the residual, are $\delta_a^{(H)} = w \odot \phi'(z_a^{(H)})$ and $\delta_a^{(\ell)} = \phi'(z_a^{(\ell)}) \odot (W^{(\ell+1)})^\top \delta_a^{(\ell+1)}$. With block mobilities $(n, 1, \dots, 1, n)$, gradient flow reads

$$\dot{W}^{(1)} = -\frac{2}{m} \sum_a r_a \delta_a^{(1)} \frac{x_a^\top}{\sqrt{d}}, \quad \dot{W}^{(\ell)} = -\frac{2}{nm} \sum_a r_a \delta_a^{(\ell)} (h_a^{(\ell-1)})^\top \quad (2 \leq \ell \leq H), \quad \dot{w} = -\frac{2}{m} \sum_a r_a h_a^{(H)}. \tag{2}$$

We write $\theta = (W^{(1)}, \dots, W^{(H)}, w)$ for all parameters and $\dot{\theta} = F(\theta)$ for (2). Hidden entries are initialized as $\mathcal{N}(0, 1/n)$, first-layer entries as $\mathcal{N}(0, 1)$, and the stored readout is small.

Population flow. As $n \rightarrow \infty$, the neurons of layer ℓ become a probability space Ω_ℓ with expectation $\mathbb{E}_\ell$: hidden coordinates $z^{(\ell)}, h^{(\ell)}$ become random variables in $L^2(\Omega_\ell)$, the first-layer weights a random row $W^{(1)} \in L^2(\Omega_1; \mathbb{R}^d)$, the readout $w \in L^2(\Omega_H)$, and each hidden matrix a bounded operator $W^{(\ell)} : L^2(\Omega_{\ell-1}) \rightarrow L^2(\Omega_\ell)$ used together with its true adjoint. The same equations (2) hold with $\frac{1}{n} \sum_i$ replaced by $\mathbb{E}_\ell$ and outer products $\delta h^\top / n$ by the rank-one operators $g \mapsto \delta \mathbb{E}[h g]$. Such limits are established in several scopes—for fixed depth on a short horizon, and globally in special models—with finite-width gradient flow converging to them on compact horizons; a global theorem for general data and depth is open. Where we need the population flow we state its existence as part of the model.

4 Compressing a dynamical system

Exact and approximate; local and global. An exact compression reproduces the reference trajectory; an approximate one reproduces it up to a controllable error. A compression may also match the current state and velocity while its small errors compound until the trajectories separate; we ask for the stronger, global notion.

Definition 4.1 (Global approximation on compact horizons). Autonomous systems indexed by an order N , with trajectories θ_N , *globally approximate* a reference flow θ if for every $T < \infty$

$$\sup_{0 \leq t \leq T} \|\theta_N(t) - \theta(t)\| \longrightarrow 0 \quad (N \rightarrow \infty).$$

The rate in N quantifies the trade-off between the size of the compressed state and its accuracy.

Three classes of state.

1. A *scalar ODE* evolves finitely many real numbers, independently of width and training time.
2. A *population* (or field) ODE evolves finitely many functions, random variables or probability laws on fixed spaces—for example the law of a neuron’s weights. Its particle implementation stores one value per particle and field; the number of objects is finite, their degrees of freedom are not.
3. An *operator* ODE evolves operators between population spaces, the limits of matrices. A matrix records how *each* row relates to *each* column, and this row–column coupling cannot in general be captured by finitely many populations.

A finite number of named fields or operators need not contain finitely many scalars, so every compression claim below names both the type and the size of the retained state.

5 Exact compression: evidence that it lives in the operator class

Before allowing errors we ask what exact capture of the population flow can look like. We do not settle this question; we collect three pieces of evidence that point the same way.

An exact operator description exists, and is finite only in name. If we refuse the growing histories of Tensor Programs and DMFT, the natural exact description of the population flow is an operator ODE: (2) read with the hidden matrices $W^{(\ell)}$ as operators. In a deep linear benchmark—one input, three linear hidden layers, order-one initialization—this can be proved completely: the learned part of the whole network becomes a single trace-class operator that evolves by an explicit autonomous cubic equation driven by the residual, and finite gradient flow and gradient descent converge to it on every compact horizon (Appendix D). The description is exact and consists of one operator, but that operator is as rich as the learned matrix: any finite approximation of it essentially stores either the dense matrix or the responses that built it.

The past can be dropped, but only for an infinite hierarchy. For a two-hidden-layer population flow on the circle (Appendix C), one can describe the dynamics exactly through *current* observations: joint laws, on each neuron population, of finitely many expressions built from the current weights, the initialization, ϕ , bounded products and applications of $W^{(2)}$ and its adjoint. The derivative of each finite level of this hierarchy is expressed through a finite higher level, and the complete hierarchy determines predictions on the whole input space and a unique continuation from every reachable state, without storing the past. But no finite level closes.

Exact scalar closures fail. In the same linear benchmark, no encoder formed from a fixed finite list of tensor contractions of the weights with width-independent bounded degree admits a polynomial autonomous scalar ODE and polynomial readout that reproduce the output dynamics on an open set of states at all large widths (Appendix D). The reason is that repeated differentiation of the output forces ever larger connected contractions: the $(2j - 1)$ st derivative contains

$$\frac{1}{n} (W^{(1)})^\top \left((W^{(2)})^\top (W^{(3)})^\top W^{(3)} W^{(2)} \right)^j W^{(1)},$$

of degree $4j + 2$, which no bounded-degree alphabet can express. The result is restricted to a natural class of encoders, but if exact scalar closure fails for three *linear* layers, there is little reason to expect it for nonlinear feature learning.

Between the two. Whether the exact operator can be compressed into finitely many populations is open. We believe not: natural reductions of an operator to populations assume structure, such as finite rank, that feature learning does not provide, since gradient flow integrates rank-one updates over continuous time. Taken together, the evidence suggests that exact capture of deep nonlinear feature learning is possible only in the operator class, which hides a growing complexity in finite clothing. The rest of the paper asks what can be gained by accepting a controlled error.

6 Approximate compression with rates: response memories

We now ask what can be gained by accepting a controlled error, for networks of any depth and with explicit order-versus-error rates.

6.1 The idea

Integrating (2), every hidden matrix is

$$W^{(\ell)}(t) = W_0^{(\ell)} - \frac{2}{nm} \sum_a \int_0^t r_a(s) \delta_a^{(\ell)}(s) (h_a^{(\ell-1)}(s))^\top ds. \quad (3)$$

The learned part is an integral of backward against forward responses. Storing it means storing the dense matrix or the whole response history. But to predict where a moving ball goes next we need its position, velocity and acceleration, not its past positions. If each response history is regular in time, a few coefficients should summarize it. Two things must hold: the summary must be *closed*, so that it evolves autonomously, and *accurate* over the whole of training, which alternates between bursts of learning and plateaus.

6.2 Closure: shifted Legendre moments

Let p_k be the shifted Legendre polynomials on $[0, 1]$, normalized by $p_k(1) = 1$ and $\int_0^1 p_j p_k = \mathbf{1}_{j=k}/(2k + 1)$. Their derivatives reduce to lower modes,

$$x p'_k(x) = k p_k(x) + \sum_{j < k} (2j + 1) p_j(x), \quad (4)$$

and this is what makes the moment system close exactly. The identity underlies online polynomial memory in HiPPO (Gu et al., 2020); the new element is that the memorized signals are the network’s own forward and backward responses, coupled back through the reconstructed network.

6.3 Accuracy: a clock that follows learning

A polynomial basis is accurate only if history is spread evenly over its window, and in physical time it is not. We therefore measure history in a clock τ that runs at the speed of learning,

$$\dot{\tau} = \rho, \quad \tau(0) = 1 : \quad (5)$$

fast while the network learns, slow as learning stalls. The unit prefix $[0, 1]$ carries the initial forward responses and no backward response. For each hidden link $\ell \geq 2$ and sample a we store the first P Legendre moments, in the clock coordinate, of the forward response $h_a^{(\ell-1)}$ and of the residual-weighted backward response $r_a \delta_a^{(\ell)} / \rho$:

$$\bar{h}_{a,k}^{(\ell-1)} = \int_0^\tau h_a^{(\ell-1)} p_k(\xi/\tau) d\xi, \quad \bar{\delta}_{a,k}^{(\ell)} = \int_0^\tau \frac{r_a \delta_a^{(\ell)}}{\rho} p_k(\xi/\tau) d\xi, \quad 0 \leq k < P,$$

where ξ is the clock value at which each response was recorded. By (4) these moments obey the closed system

$$\begin{aligned} \dot{\bar{h}}_{a,k}^{(\ell-1)} &= \rho h_a^{(\ell-1)} - \frac{\rho}{\tau} \left[k \bar{h}_{a,k}^{(\ell-1)} + \sum_{j < k} (2j+1) \bar{h}_{a,j}^{(\ell-1)} \right], \\ \dot{\bar{\delta}}_{a,k}^{(\ell)} &= r_a \delta_a^{(\ell)} - \frac{\rho}{\tau} \left[k \bar{\delta}_{a,k}^{(\ell)} + \sum_{j < k} (2j+1) \bar{\delta}_{a,j}^{(\ell)} \right], \end{aligned} \quad (6)$$

with $\bar{h}_{a,0}^{(\ell-1)}(0) = h_a^{(\ell-1)}(0)$ and all other moments zero, and the hidden matrices are reconstructed as

$$\widehat{W}^{(\ell)} = W_0^{(\ell)} - \frac{2}{nm\tau} \sum_{a=1}^m \sum_{k=0}^{P-1} (2k+1) \bar{\delta}_{a,k}^{(\ell)} (\bar{h}_{a,k}^{(\ell-1)})^\top. \quad (7)$$

All responses in (6) are evaluated in the current *reconstructed* network, and the first layer and readout follow (2). The system is autonomous and never divides by ρ . Equivalently, with Π_P the $L^2(0, \tau)$ projection onto polynomials of degree below P ,

$$\widehat{W}^{(\ell)} = W_0^{(\ell)} - \frac{2}{nm} \sum_a \int_0^\tau \left(\Pi_P \frac{r_a \delta_a^{(\ell)}}{\rho} \right) (\Pi_P h_a^{(\ell-1)})^\top d\xi, \quad (8)$$

and since cross terms vanish by orthogonality, the error in each matrix update is a *product* of a forward and a backward projection error.

The joint feature-learning clock. The learning-speed clock makes forward histories smooth but not every backward history. A second clock advances with the residual *plus* the speed of all forward and backward responses,

$$\dot{\tau} = \rho + \left\| \frac{d}{dt} (h_a^{(\ell-1)}, r_a \delta_a^{(\ell)} / \rho)_{\ell,a} \right\|_2, \quad \tau(0) = 1, \quad (9)$$

so that the whole response path moves at unit speed at most in the clock coordinate. History is placed by this clock but still weighted by the residual mass ρdt , so the Legendre modes are no longer orthogonal and one shared $P \times P$ Gram matrix is tracked alongside the moments; the reconstruction uses its inverse, the defect keeps the product form, and the clock speed can be evaluated explicitly by one directional forward and backward pass (Appendix B).

6.4 Global guarantees

Theorem 6.1 (Learning-speed clock). *Fix finite width, depth $H \geq 2$, arbitrary finite data and a finite initialization, and let every activation be in $C_{\text{loc}}^{1,1}(\mathbb{R})$. For every $T < \infty$ there are $C(T)$ and $P_0(T)$, independent of P , such that for $P \geq P_0(T)$ the closure exists uniquely on $[0, T]$ and*

$$\sup_{0 \leq t \leq T} \left\| \hat{\theta}_P(t) - \theta(t) \right\| \leq \frac{C(T)}{\sqrt{P(P+1)}}.$$

The rate extends to hidden responses and to predictions on any bounded set of unseen inputs. If ϕ and ϕ' are bounded (tanh, sigmoid, arctangent, sine), it holds for every $P \geq 1$. No assumption of successful fitting, Gaussian initialization, input geometry or label structure is made.

Theorem 6.2 (Joint feature-learning clock). *Under the setting of Theorem 6.1, assume the activations of layers $\ell \geq 2$ are in $C_{\text{loc}}^{2,1}$ and $\rho(0) > 0$. For every T there are $P_0(T)$ and $C(T)$ such that for $P \geq P_0(T)$ the closure exists uniquely on $[0, T]$, its clock stays bounded independently of P , and $\sup_{t \leq T} \left\| \hat{\theta}_P(t) - \theta(t) \right\| \leq C(T)/[P(P+1)]$.*

What separates this from approximating a given signal, as in HiPPO, is feedback: the histories being memorized are produced by the network reconstructed from the memories, so approximation errors change the signals they approximate. Both proofs (Appendices A and B) rest on three facts: an exact identity for the growth of the projection error of each history; the product structure of the defect; and the Legendre tail bound $\|v - \Pi_P v\|^2 \leq \tau^2 \|v'\|^2 / (4P(P+1))$. With the learning-speed clock the forward histories have P -independent derivative bounds while backward histories only need bounded mass, giving $O(P^{-1})$; with the joint clock the whole path is unit speed once the clock is shown to be bounded, both factors contribute, and the rate is $O(P^{-2})$. Two steps carry the difficulty. In the learning-speed clock, bounding forward histories through the layers requires the reconstructed velocity, whose memory endpoint term grows like P ; it is cancelled exactly by the P^{-1} projection factor, which is why the rate does not degrade with depth. In the joint clock, the clock itself must be bounded without assuming it, which we do through an order-independent bound on the total variation of the trajectory. Gronwall's inequality and a first-exit argument then transfer the accumulated defect to the trajectory.

Corollary 6.3 (Learned increments are low-rank in their own history). *Under Theorem 6.1, for every T and every $P \geq P_0(T)$, each dense learned increment $W^{(\ell)}(t) - W_0^{(\ell)}$, $0 \leq t \leq T$, is within $C(T)/\sqrt{P(P+1)}$ in Frobenius norm of the matrix $-\frac{2}{nm\tau} \sum_{a,k < P} (2k+1) \bar{\delta}_{a,k}^{(\ell)} (\bar{h}_{a,k}^{(\ell-1)})^\top$, which has rank at most mP and is built from the moments of the closure's own responses. Under Theorem 6.2, the joint-clock reconstruction, of rank at most $m(P+1)$, is within $C(T)/[P(P+1)]$.*

The corollary follows from the theorems because the reconstruction (7) has rank at most mP . Its content is not that learned increments admit low-rank approximations—a truncated singular value decomposition would supply those, but only from the dense matrix itself. It is that a good approximation is *generated autonomously and online* from the network's own forward and backward history, with a rank that does not grow with time: on a given horizon, one fixed order serves every time inside it, however much the features have moved. The statement is informative only while $mP < n$, and its constant depends on width.

6.5 Cost

With H layers, width n , m samples and order P , each learned $n \times n$ increment is carried by $2mP$ vectors of length n . The moving internal state is $2(H-1)mnP$ numbers against $(H-1)n^2$ for

the dense learned matrices; it does not grow with training time and is smaller whenever $2mP < n$. The initialized matrices, $(H - 1)n^2$ numbers, are kept as a fixed source. Per step, applying $W_0^{(\ell)}$ and its transpose costs $O(mn^2)$ per layer as in dense training, the learned correction $O(m^2nP)$ per layer, and the moment update $O(HmnP)$; the joint clock adds an $O(P^3)$ Gram factorization and $O(HmnP^2)$ solves. Trajectory error ε requires $O(Hmn\varepsilon^{-1})$ history numbers with the learning-speed clock and $O(Hmn\varepsilon^{-1/2} + \varepsilon^{-1})$ with the joint clock.

What is compressed is time, not samples or neurons. The state grows linearly with the number of training samples, and once $2mP \geq n$ it exceeds the dense learned matrices; learning from a continuum of inputs would need a further representation of the input dependence. Nor is this a runtime improvement: because $W_0^{(\ell)}$ still acts densely, the closure ran somewhat slower than dense training in our experiments.

6.6 From finite networks to the population

The theorems hold at every fixed width, for every realized initialization. At the population level they suggest the following picture, which we state as an interpretation, not a result. For each layer and sample, the P forward and P backward moments are random variables over neurons, and the learned action of $W^{(\ell)}$ becomes $g \mapsto -\frac{2}{m\tau} \sum_{a,k} (2k+1) \bar{\delta}_{a,k}^{(\ell)} \mathbb{E}_{\ell-1}[\bar{h}_{a,k}^{(\ell-1)} g]$: finitely many neuron-level memory fields, coupled through the initialized operator. A width- n network would then be a Monte Carlo simulation of this system. What is proved is only a diagonal statement: if finite-width flows converge to a population flow on $[0, T]$, choosing $n_j \rightarrow \infty$ and then P_j large enough at each width yields compressed systems converging to it. The natural next theorem is the following.

Conjecture 6.4 (Fixed-order population closure). *For each fixed P , as $n \rightarrow \infty$ the order- P response-memory closure converges on compact horizons to a limiting system consisting of the initialized operators and finitely many neuron-level memory fields per layer and sample; and this limit tracks population gradient flow with error $O_T(P^{-1})$, or $O_T(P^{-2})$ with the joint clock, uniformly in width.*

The limit in the conjecture would be a time-local, finite-order description of deep nonlinear feature learning in the population limit: the kind of object the introduction asks for.

7 Experiments

We integrate finite networks and compare response memory with dense gradient flow from the same initialization, using the learning-speed clock unless stated. With eight labelled points on a circle, many functions fit the labels, so agreement at training points says little; we compare the whole function that training selects, on 8192 angles, at each model’s first crossing of training MSE 0.001. The five tasks are chosen to be hard: alternating labels, clustered points and large unlabelled arcs. These are comparisons of fitted functions at matched loss, with one initialization per task, not same-time trajectory errors: they test the closure as an estimator of what dense training produces, not the rates of the theorems.

Circle tasks. In two layers the error falls by 1.5–3 orders of magnitude from $P = 1$ to $P = 7$ on every task, faster than the P^{-1} trajectory bound (Figure 2); the theorem is an upper bound on a different quantity, and we do not read a rate off these five points. In three layers $P = 2$ and $P = 3$ stay within 0.06 of the dense function everywhere, and within 0.01 on three tasks, while the hidden layers move substantially: relative activation displacement is 0.35–0.65, 0.38–0.74 and 0.46–0.68 in

Functions on the circle at matched training loss

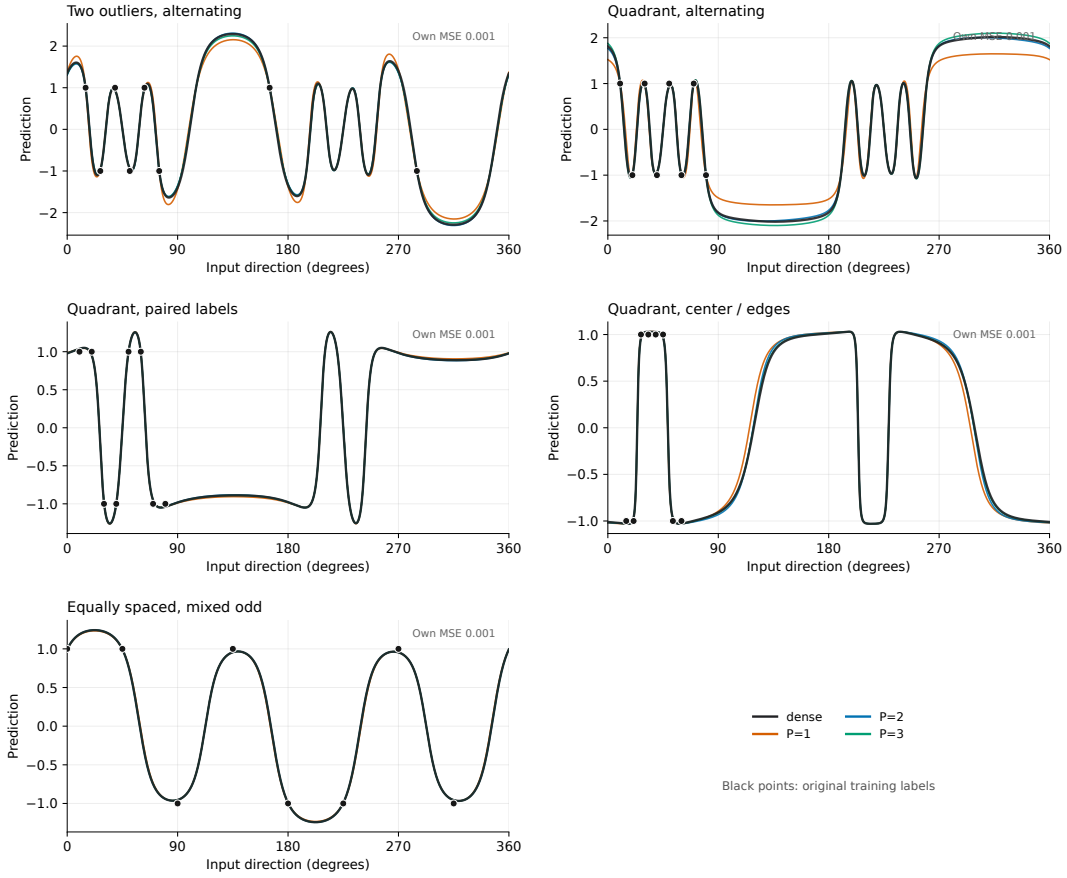


Figure 1: Three hidden tanh layers, width 4096. Black: dense gradient flow; colours: response memory of order $P = 1, 2, 3$; dots: training labels.

the three layers. Order is not monotone at fixed width: $P = 2$ beats $P = 3$ on both alternating tasks. The moving state at $P = 3$ is about 3 MiB against 256 MiB for the dense learned matrices.

Evolving memories versus fixed or trained bases. The corollary says that learned increments are low-rank *in their own history*. Other low-rank bases do not work. Frozen dictionaries built at initialization reach RMS 0.55–1.38 on the outlier task with about 45 vectors, against 0.073 for 48 response moments. A directly trained low-rank correction of the middle matrix, with factors of matched rank and the same $W_0^{(2)}$, lost in all 29 valid comparisons, 28 of them by more than a factor of two; at rank 56 on the outlier task the errors were 0.0047 against 0.52 and 0.25 for two factor seeds.

MNIST. Predictions of response memory and dense training on held-out images lie on the diagonal (Figure 3), with identical test accuracy (94.7%). With 1000 training images and width 1024 the differences are 0.019, 0.0019 and 0.0012 for $P = 1, 2, 3$, with 97.5% accuracy for dense, $P = 2$ and $P = 3$.

Table 1: Whole-circle RMS difference between response memory and dense gradient flow. Left: two hidden tanh layers, width 2048. Right: three hidden tanh layers, width 4096 (Figure 1).

Task	two layers			three layers		
	$P = 1$	$P = 3$	$P = 7$	$P = 1$	$P = 2$	$P = 3$
Two outliers, alternating	0.236	0.0730	0.00466	0.108	0.0238	0.0315
Quadrant, alternating	0.341	0.0456	0.00155	0.275	0.0284	0.0585
Quadrant, paired	0.0193	0.00282	0.00012	0.0122	0.00371	0.00121
Quadrant, center/edges	0.0456	0.00488	0.00131	0.0790	0.0257	0.00725
Equally spaced, mixed	0.0124	0.00053	4×10^{-6}	0.0150	0.00241	0.00028

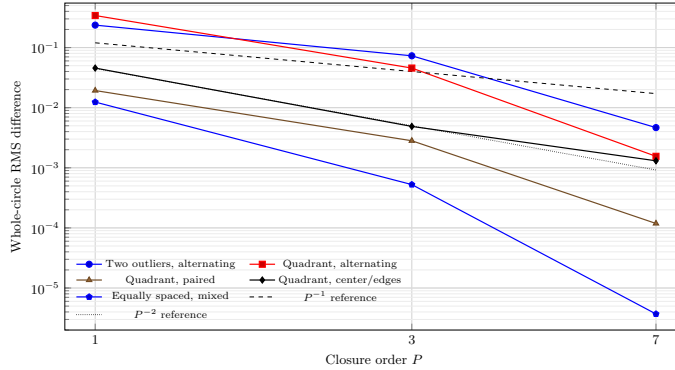


Figure 2: Whole-circle difference from dense training against order, two hidden tanh layers (left half of Table 1), with reference slopes. These are matched-loss discrepancies of fitted functions, not the same-time trajectory error bounded by Theorem 6.1.

Two clocks. The joint clock’s better rate is asymptotic. At $P = 5$ it improved some cases (tanh alternating quadrant $0.0066 \rightarrow 0.0035$; GELU outliers $0.0020 \rightarrow 0.0010$) and was much worse on another (GELU alternating quadrant $0.013 \rightarrow 1.00$, reduced to 0.10 by halving the step). Over an 82-configuration sweep, on the 65 configurations where dense and all orders fit, the joint clock’s median differences were 0.031, 0.013 and 0.0066 for $P = 1, 2, 3$.

8 Discussion: toward an idealized model of learning

We asked whether the idealized limit of deep feature learning can be described by a small state, in the way a large physical system is described by a few collective variables. Exactly, the evidence says no: autonomous exact descriptions appear to require operators or infinite hierarchies, and simple scalar closures fail. Approximately, a different kind of compression appears. The known exact descriptions compress width and keep the whole history; we keep the neurons and compress the history. Measured in a clock that follows learning, the memory of deep nonlinear feature learning is summarized by a few moments per neuron and sample, with an error that decreases polynomially in the number of moments and a state that does not grow with training time. The closest historical analogue is the moment closure of kinetic theory, where a few moments of a distribution replace its full evolution; such closures are usually adopted without convergence guarantees, whereas here the truncation comes with a rate.

The variables that survive are *neuron-level memory variables*: for each neuron and sample, compressed summaries of how it has responded forward and backward, coupled through a fixed

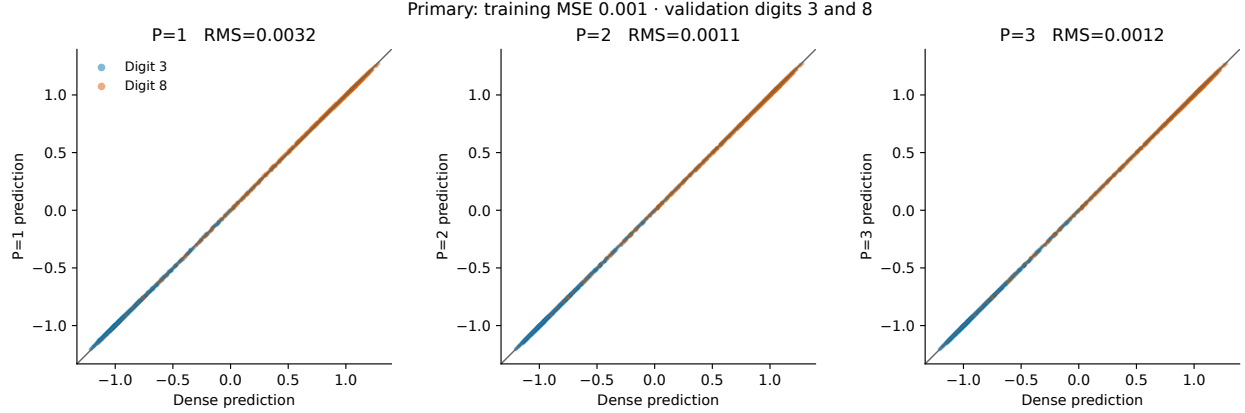


Figure 3: MNIST 3 versus 8, 100 training images, width 4096: response memory versus dense predictions on 1984 held-out images. RMS differences are 0.0032, 0.0011 and 0.0012 for $P = 1, 2, 3$.

random operator. They become collective only at the population level, and Conjecture 6.4 states what that would require. Other gaps remain. The initialized random operator W_0 is kept exactly; it plays the role of a fixed random environment, and compressing it while preserving its forward and adjoint reuse is the main open problem. The constants depend on width and horizon, and the small orders that work in practice are not explained by the asymptotic rates. We have not ablated the clock against physical time. Finally, the theory covers smooth activations, and the memory grows with the number of training samples, so learning over a continuum of inputs needs a further representation of the input dependence.

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A Proof of Theorem 6.1

Throughout the appendices we write $b_a^{(\ell)} = r_a \delta_a^{(\ell)} / \rho$ for the residual-weighted backward response, $\bar{h}_a^{(\ell-1)}$ and $\bar{b}_a^{(\ell)}$ for its and the forward response’s histories in the clock coordinate ξ (extended over the unit prefix as in the main text), and $h_a^{(\ell-1)*}$, $b_a^{(\ell)*}$ for the values at the current endpoint of their degree- $(P-1)$ Legendre projections. The only approximation-theoretic input is the following Legendre tail bound.

Lemma A.1 (Legendre tail). *For $q \in H^1(0, \tau; \mathbb{R}^N)$,*

$$\|q - \Pi_P q\|_{L^2(0, \tau)}^2 \leq \frac{1}{P(P+1)} \int_0^\tau \xi(\tau - \xi) \|q'(\xi)\|^2 d\xi \leq \frac{\tau^2}{4P(P+1)} \|q'\|_{L^2(0, \tau)}^2. \quad (10)$$

Proof. On $[0, 1]$, the shifted Legendre equation is

$$-\frac{d}{dx} \{x(1-x)p'_k(x)\} = k(k+1)p_k(x).$$

Integration by parts shows that the derivatives are orthogonal in the weight $x(1-x)$ and have squared norm $k(k+1)/(2k+1)$. Applying Bessel’s inequality to q' in this weighted space gives

$$\sum_{k \geq 1} k(k+1) \frac{\|c_k\|^2}{2k+1} \leq \int_0^1 x(1-x) \|q'(x)\|^2 dx,$$

where $c_k = (2k+1) \int_0^1 q(x) p_k(x) dx$. Parseval, polynomial density and $k(k+1) \geq P(P+1)$ for $k \geq P$ give the first bound on $[0, 1]$. Rescaling and $\xi(\tau - \xi) \leq \tau^2/4$ give (10); summing componentwise proves the vector statement. $\square$

Step 1: dense existence and a common compact region. The mobility matrix $\mathcal{D}$ has largest eigenvalue n , and $F = -\mathcal{D}\nabla\mathcal{L}$. Hence

$$\dot{\mathcal{L}} = -\left\|\mathcal{D}^{-1/2}\dot{\theta}\right\|^2, \quad \int_0^t \left\|\dot{\theta}(s)\right\|^2 ds \leq n\rho(0)^2. \quad (11)$$

Thus $\|\theta(t) - \theta(0)\| \leq \sqrt{nt}\rho(0)$. A finite maximal endpoint would have a finite limiting state, from which local existence would extend the solution, so the dense flow is global.

Now fix T and let

$$R = \|\theta(0)\| + \sqrt{nT}\rho(0) + 1, \quad \mathcal{B} = \{\vartheta : \|\vartheta\| \leq R\}.$$

The dense path stays at distance at least one from $\partial\mathcal{B}$. We stop the closure at its first exit from $\mathcal{B}$; the rest of the proof uses constants on this explicit ball and eventually rules out that exit.

To make the constants concrete, set $X = \max_a \|x_a\|/\sqrt{d}$ and $\alpha_0 = X$, and choose recursively

$$\begin{aligned} s_\ell &= \sup_{|z| \leq R\alpha_{\ell-1}} |\phi'(z)|, & \alpha_\ell &= \sqrt{n} \sup_{|z| \leq R\alpha_{\ell-1}} |\phi(z)|, \\ \beta_H &= Rs_H, & \beta_\ell &= s_\ell R\beta_{\ell+1} \quad (\ell < H). \end{aligned} \quad (12)$$

Then $\|h_a^{(\ell)}\| \leq \alpha_\ell$ and $\|\delta_a^{(\ell)}\| \leq \beta_\ell$ on $\mathcal{B}$. Put

$$\begin{aligned} Y &= \left(m^{-1} \sum_a y_a^2\right)^{1/2}, & q &= R\alpha_H/n + Y, & S &= Tq, & A &= 1 + S, \\ v_1 &= 2\beta_1 X, & v_\ell &= 2\beta_\ell \alpha_{\ell-1}/n, & v_w &= 2\alpha_H, \\ V &= \left(v_1^2 + \sum_{\ell=2}^H v_\ell^2 + v_w^2\right)^{1/2}. \end{aligned} \quad (13)$$

On the stopped interval, $\rho \leq q$, $\tau \leq A$ and $\|F\| \leq V\rho$. Since F is locally Lipschitz, it has some finite Lipschitz constant K on $\mathcal{B}$.

Step 2: an exact projection-error energy. For any single forward or backward history q , let $D_q(t) = \|q - \Pi_P q\|_{L^2(0, \tau(t))}^2$, and let q^* be the polynomial projection evaluated at the current endpoint. Differentiating the raw energy and the projected energy with (6), every basis-dilation term cancels, leaving

$$\dot{D}_q = \rho \|q - q^*\|^2, \quad D_q(t) = \int_0^t \rho(s) \|q(s) - q^*(s)\|^2 ds. \quad (14)$$

The jump at the end of the backward prefix is harmless, since only square integrability and growing integrals are used.

Next differentiate (7). The physical closure equation is exactly

$$\dot{\hat{\theta}}_P = F(\hat{\theta}_P) + E, \quad E_1 = E_w = 0, \quad (15)$$

with internal defects

$$E_\ell = \frac{2\rho}{nm} \sum_a (b_a^{(\ell)} - b_a^{(\ell)*})(h_a^{(\ell-1)} - h_a^{(\ell-1)*})^\top. \quad (16)$$

The two mixed projection terms vanish by orthogonality, so the defect is a pure product of errors. Consequently

$$\int_0^t \|E_\ell(s)\|_F ds \leq \frac{2}{nm} \sum_a \sqrt{D_{b,\ell,a}(t) D_{h,\ell,a}(t)}. \quad (17)$$

This inequality is the bridge from a same-history approximation bound to a trajectory theorem. Define also the proof-only accumulator

$$W_{\text{acc}}^{(\ell)}(t) = W_0^{(\ell)} - \frac{2}{nm} \sum_a \int_0^t \hat{r}_a \hat{\delta}_a^{(\ell)} (\hat{h}_a^{(\ell-1)})^\top ds, \quad R_\ell = \widehat{W}^{(\ell)} - W_{\text{acc}}^{(\ell)}.$$

Then $R_\ell(0) = 0$ and $\dot{R}_\ell = E_\ell$, so (17) controls the accumulated absolute velocity error, not only a signed endpoint integral.

Step 3: forward histories have a P -independent derivative bound. Before changing variables, note that a nonstationary learning-speed solution cannot first reach $\rho = 0$ at a finite regular state. Every raw velocity vanishes there, and local uniqueness applied backward from that equilibrium would force the preceding path to be stationary. Hence $\xi = \tau(t)$ is a valid coordinate on every nonstationary compact segment; the raw algorithm itself remains defined at $\rho = 0$ without division.

Write primes for derivatives in $\xi = \tau(t)$ and define

$$Z_\ell(t) = \frac{1}{m} \sum_a \int_0^{\tau(t)} \left\| (h_a^{(\ell)})'(\xi) \right\|^2 d\xi.$$

The first layer satisfies $\left\| (h_a^{(1)})' \right\| \leq s_1 v_1 X$. For later layers, the chain rule gives

$$(h_a^{(\ell)})' = \phi'(z_a^{(\ell)}) \odot \left[(F_\ell/\rho + E_\ell/\rho) h_a^{(\ell-1)} + \widehat{W}^{(\ell)} (h_a^{(\ell-1)})' \right]. \quad (18)$$

The delicate term is the defect, because endpoint evaluation of a polynomial projection can amplify errors by a factor that grows with P . Endpoint evaluation of a degree- $(P-1)$ polynomial has operator norm $P/\sqrt{\tau}$ from $L^2(0, \tau)$, because $\sum_{k < P} (2k+1) = P^2$. Projection contraction and $m^{-1} \sum_a \left\| b_a^{(\ell)} \right\|^2 \leq \beta_\ell^2$ then imply

$$\left(m^{-1} \sum_a \left\| b_a^{(\ell)*} \right\|^2 \right)^{1/2} \leq P\beta_\ell, \quad \left(m^{-1} \sum_a \left\| b_a^{(\ell)} - b_a^{(\ell)*} \right\|^2 \right)^{1/2} \leq (P+1)\beta_\ell.$$

Combining this with Lemma A.1, (14) and (16) yields

$$\int_0^t \rho \|E_\ell/\rho\|_F^2 ds \leq \frac{2\beta_\ell^2 \mathbf{A}^2}{n^2} Z_{\ell-1}(t). \quad (19)$$

Here the endpoint factor $P+1$ cancels the P^{-1} projection factor. This cancellation is the step that prevents a loss of convergence order with depth.

Define finite, P -independent bounds recursively by

$$\begin{aligned}\bar{Z}_1 &= S(s_1 v_1 X)^2, \\ \bar{Z}_\ell &= 3s_\ell^2 \left[S v_\ell^2 \alpha_{\ell-1}^2 + \left(R^2 + \frac{2\beta_\ell^2 A^2 \alpha_{\ell-1}^2}{n^2} \right) \bar{Z}_{\ell-1} \right].\end{aligned}\tag{20}$$

Integrating (18), and using $\|x + y + z\|^2 \leq 3(\|x\|^2 + \|y\|^2 + \|z\|^2)$ together with (19), proves inductively that $Z_\ell \leq \bar{Z}_\ell$.

Step 4: accumulated defect and feedback. The zero backward prefix gives $m^{-1} \sum_a D_{b,\ell,a} \leq S\beta_\ell^2$, and Lemma A.1 gives

$$\frac{1}{m} \sum_a D_{h,\ell,a} \leq \frac{A^2 \bar{Z}_{\ell-1}}{4P(P+1)}.$$

Summing (17) over links, we obtain

$$\int_0^t \|E(s)\| \, ds \leq \frac{B_{\text{LS}}}{\sqrt{P(P+1)}}, \quad B_{\text{LS}} = \frac{A}{n} \sum_{\ell=2}^H \beta_\ell \sqrt{S \bar{Z}_{\ell-1}}.\tag{21}$$

It remains to account for feedback. Subtracting the dense flow from the perturbed flow and applying the Lipschitz bound on F gives

$$e(t) \leq \frac{B_{\text{LS}}}{\sqrt{P(P+1)}} + K \int_0^t e(s) \, ds,$$

and Gronwall's inequality yields

$$\sup_{t \leq T} e(t) \leq \frac{B_{\text{LS}} e^{KT}}{\sqrt{P(P+1)}}.\tag{22}$$

Choose P_0 so that the right side is less than $1/2$. The closure then cannot reach $\partial\mathcal{B}$. Its moment representations, $1 \leq \tau \leq A$ and the response bounds place its whole finite state in a compact subset of the locally Lipschitz domain. A finite maximal endpoint would therefore extend, so the closure exists through T . Lipschitz dependence of the network responses on θ gives the final prediction statement.

Globally bounded activations. Suppose now that every activation and its first derivative are globally bounded. Set

$$\alpha_\ell = \sqrt{n} \|\phi\|_\infty, \quad s_\ell = \|\phi'\|_\infty, \quad B_w = \sqrt{\|w(0)\|^2 + nY^2 T}.$$

For both the dense and the closure flows, the unchanged readout equation obeys

$$\frac{d}{dt} \|w\|^2 = nY^2 - 4n \frac{1}{m} \sum_a (f_a - y_a/2)^2 \leq nY^2.$$

Therefore $\|w\| \leq B_w$, $\rho \leq q := B_w \alpha_H / n + Y$, $S = Tq$ and $\tau \leq A := 1 + S$. Start with $\beta_H = B_w s_H$ and proceed downward: for $\ell = H, \dots, 2$, define

$$D_\ell = \left\| W_0^{(\ell)} \right\|_F + \frac{2A\beta_\ell \alpha_{\ell-1}}{n}, \quad \beta_{\ell-1} = s_{\ell-1} D_\ell \beta_\ell.$$

Projection contraction in (8) gives $\|\widehat{W}^{(\ell)}\|_F \leq D_\ell$, and the backward recurrence then gives $\|\delta^{(\ell-1)}\| \leq \beta_{\ell-1}$. There is no circularity, because the descent starts from the bounded readout. Finally,

$$\|W^{(1)}(t)\|_F \leq \|W^{(1)}(0)\|_F + 2SX\beta_1.$$

These P -independent physical bounds, the moment integral representations and $1 \leq \tau \leq A$ place the order- P state in a compact set for every fixed P , so the closure continues through T for every $P \geq 1$. Repeating Steps 2–4 with these global constants gives (22) for every P . This completes the proof. $\square$

B The joint feature-learning clock: construction and proof of Theorem 6.2

Construction. Let Ψ be the concatenation over $\ell \geq 2$ and a of the forward and residual-weighted backward responses $(h_a^{(\ell-1)}, b_a^{(\ell)})$, so that the clock (9) reads $\dot{\tau} = g = \rho + \|\dot{\Psi}\|_2$ and $\|\mathrm{d}\Psi/\mathrm{d}\tau\| \leq 1$. History is placed at $\xi = \tau(s)$ with mass $\rho(s)\mathrm{d}s$, while the prefix $[0, 1]$ carries Lebesgue mass and constant initial responses; the resulting measure μ_t satisfies $\mathrm{d}\mu_t \leq \mathrm{d}\xi$. With $p = (p_0, \dots, p_{P-1})^\top$ evaluated at ξ/τ , $e = (1, \dots, 1)^\top$, and the triangular matrix $\mathsf{T}_{kk} = k$, $\mathsf{T}_{kj} = 2j + 1$ for $j < k$ (zero above the diagonal), the state consists of the moment matrices $\bar{h}_a^{(\ell-1)} = \int \bar{h}_a^{(\ell-1)}(\xi) p^\top \mathrm{d}\mu_t$, $\bar{\delta}_a^{(\ell)} = \int \bar{b}_a^{(\ell)}(\xi) p^\top \mathrm{d}\mu_t$ and one shared Gram matrix $G = \int pp^\top \mathrm{d}\mu_t$, which evolve by

$$\begin{aligned} \widehat{W}^{(\ell)} &= W_0^{(\ell)} - \frac{2}{nm} \sum_a (\bar{\delta}_a^{(\ell)} G^{-1} (\bar{h}_a^{(\ell-1)})^\top - b_a^{(\ell)}(0) h_a^{(\ell-1)}(0)^\top), \\ \dot{\bar{h}}_a^{(\ell-1)} &= \rho h_a^{(\ell-1)} e^\top - (g/\tau) \bar{h}_a^{(\ell-1)} \mathsf{T}^\top, \quad \dot{\bar{\delta}}_a^{(\ell)} = r_a \delta_a^{(\ell)} e^\top - (g/\tau) \bar{\delta}_a^{(\ell)} \mathsf{T}^\top, \\ \dot{G} &= \rho e e^\top - (g/\tau) (\mathsf{T}G + G\mathsf{T}^\top), \end{aligned} \tag{23}$$

with $G(0) = \text{diag}(1, \frac{1}{3}, \dots, \frac{1}{2P-1})$, only the zeroth moment columns nonzero initially, and the prefix product subtracted so that $\widehat{W}^{(\ell)}(0) = W_0^{(\ell)}$. With $h_a^{(\ell-1)*} = \bar{h}_a^{(\ell-1)} G^{-1} e$ and $b_a^{(\ell)*} = \bar{\delta}_a^{(\ell)} G^{-1} e$, differentiating the reconstruction cancels every g/τ term and gives the defect $E_\ell = \frac{2\rho}{nm} \sum_a (b_a^{(\ell)} - b_a^{(\ell)*})(h_a^{(\ell-1)} - h_a^{(\ell-1)*})^\top$, of the same product form as (16).

Step 1: a lower bound on the dense residual. Assume $\rho_0 > 0$ as stated, and use the compact physical ball $\mathcal{B}$ from Appendix A. Retain the constants $\alpha_\ell, \beta_\ell, q, S, A, V, K$ from the proof of Theorem 6.1. On this ball the output map has a finite Lipschitz constant c_f : prediction and residual RMS change by at most $c_f \|\theta - \theta'\|$. Along dense flow,

$$|\dot{\rho}_{\text{dense}}| \leq c_f \|F(\theta)\| \leq c_f V \rho_{\text{dense}}.$$

Consequently

$$\rho_{\text{dense}}(t) \geq \mu := \rho_0 e^{-c_f V T} > 0, \quad 0 \leq t \leq T. \tag{24}$$

Step 2: an exact weighted projection energy. Consider the maximal regular closure interval stopped before either its physical state leaves $\mathcal{B}$ or its residual reaches $\mu/2$. We first derive a clock bound on this stopped interval without assuming small projection error. For any inserted vector history f , let

$$D_f = \|f - \Pi_t f\|_{L^2(\mu_t)}^2 = \int \|f\|^2 \mathrm{d}\mu_t - \text{tr}(B_f G^{-1} B_f^\top).$$

Differentiating $G^{-1}G = I$ gives

$$\frac{d}{dt}G^{-1} = -\rho G^{-1}ee^\top G^{-1} + (g/\tau)(G^{-1}\mathsf{T} + \mathsf{T}^\top G^{-1}).$$

Together with (23), this cancels every dilation term and yields

$$\dot{D}_f = \rho \|f - f^*\|^2, \quad D_f(t) = \int_0^t \rho \|f - f^*\|^2 ds. \quad (25)$$

The initial error is zero because every prefix is constant. Thus (17) holds with the weighted errors in (25).

Step 3: order-independent coarse bounds. Put

$$Q = m \sum_{\ell=2}^H (\alpha_{\ell-1}^2 + \beta_\ell^2), \quad C_0 = \frac{AQ}{nm}.$$

The total mass of μ_t is at most $A = 1 + S$. Projection error is no larger than raw history energy, including the matching prefixes. Applying $2\sqrt{uv} \leq u + v$ in the defect estimate gives the order-independent bounds

$$\int_0^t \|E\| ds \leq C_0, \quad \int_0^t \|\dot{\hat{\theta}}\| ds \leq VS + C_0. \quad (26)$$

The second inequality uses $\dot{\hat{\theta}} = F(\hat{\theta}) + E$ and $\int_0^t \|F\| ds \leq V \int_0^t \rho ds \leq VS$. This total-variation estimate is stronger than bounded physical position, and it is obtained before any clock bound.

Step 4: a clock bound without a cap. It remains to control the derivative of the normalized response map. Let $q_r = r/\rho \in \mathbb{R}^m$. Then

$$\|q_r\|^2 = m, \quad Dq_r[v] = \left(I - \frac{q_r q_r^\top}{m} \right) \frac{Dr[v]}{\rho}. \quad (27)$$

The matrix in parentheses is an orthogonal projector, and $\rho \geq \mu/2$ on the stopped interval. Differentiating the forward and backward recurrences uses only bounded physical weights and bounded first two activation derivatives on compact preactivation intervals; the first layer uses only ϕ' . Hence there is a finite, P -independent constant J such that

$$\|D\Psi(\vartheta)\|_{\text{op}} \leq J$$

throughout the stopped tube. The stated local smoothness also makes this derivative locally Lipschitz. Since $\dot{\Psi} = D\Psi(\hat{\theta})\dot{\hat{\theta}}$, (26) now gives

$$\tau_P(t) = 1 + \int_0^t (\rho + \|\dot{\Psi}\|) ds \leq A + J(VS + C_0) =: \Lambda_T. \quad (28)$$

No clock cap has been assumed or inserted.

Step 5: the sharp joint approximation estimate. We can now use the sharper approximation estimate. The entire concatenated history $\bar{\Psi}(\xi)$ is constant on the prefix, continuous at its join and, on the physical history,

$$\|\partial_\xi \bar{\Psi}\| = \frac{\|\dot{\Psi}\|}{\rho + \|\dot{\Psi}\|} \leq 1.$$

Therefore $\int_0^{\tau_P} \|\bar{\Psi}'\|^2 d\xi \leq \tau_P - 1$. The weighted projection is a best approximation in $L^2(\mu_t)$, and $d\mu_t \leq d\xi$, so comparison with the ordinary Legendre projection and Lemma A.1 give the joint bound

$$\sum_{\ell,a} (D_{h,\ell,a} + D_{b,\ell,a}) \leq \frac{\tau_P^2(\tau_P - 1)}{4P(P+1)} \leq \frac{\Lambda_T^2(\Lambda_T - 1)}{4P(P+1)}. \quad (29)$$

This is one estimate for the concatenated path; it does not introduce a separate dimension factor per unit or layer.

Step 6: feedback and continuation. Combining (17), (28), (29) and $2\sqrt{uv} \leq u + v$ yields

$$\int_0^t \|E\| ds \leq \frac{\Gamma_{\text{JF}}}{P(P+1)}, \quad \Gamma_{\text{JF}} = \frac{\Lambda_T^2(\Lambda_T - 1)}{4nm}. \quad (30)$$

Subtracting the dense flow and applying the Lipschitz constant K and Gronwall's inequality gives

$$\|\hat{\theta}_P(t) - \theta(t)\| \leq \frac{\Gamma_{\text{JF}} e^{KT}}{P(P+1)}. \quad (31)$$

Choose $P_0(T)$ so that, for every $P \geq P_0(T)$,

$$\frac{\Gamma_{\text{JF}} e^{KT}}{P(P+1)} \leq \frac{1}{2}, \quad c_f \frac{\Gamma_{\text{JF}} e^{KT}}{P(P+1)} \leq \frac{\mu}{4}. \quad (32)$$

The first inequality rules out physical exit, because the dense path stays at distance one from $\partial\mathcal{B}$. The second, together with (24), gives $\hat{\rho} \geq 3\mu/4 > \mu/2$, ruling out residual exit.

Finally, the matching prefix supplies, at every fixed P ,

$$G(t) \succeq \int_0^1 p(\xi/\tau_P(t)) p(\xi/\tau_P(t))^\top d\xi. \quad (33)$$

The right side is positive definite, since a zero quadratic form would make a nonzero polynomial vanish on an interval. Its entries vary continuously with τ_P , so on $1 \leq \tau_P \leq \Lambda_T$ its least eigenvalue has a positive minimum $\gamma_{P,T}$; this minimum need not be uniform in P . The moment integral representations, finite history mass, bounded responses and bounded fixed-order polynomials then bound every raw coordinate. Together with the physical, residual, clock and Gram bounds, the full state lies in a compact subset of its locally Lipschitz domain. A finite maximal endpoint therefore extends, which proves continuation through T and completes the theorem with $C_T = \Gamma_{\text{JF}}$ and $K_T = K$. $\square$

Implementation of the joint clock

Equation (9) appears implicit only because $\dot{\Psi}$ is written compactly; no nonlinear solve for g is needed. The internal velocities do not depend on g , so they can be computed first. Begin with $q = G^{-1}e$, the endpoint fits a^*, b^* , and the internal velocities

$$V_\ell = \dot{\widehat{W}}^{(\ell)} = -\frac{2\rho}{nm} \sum_a \left[b_a^{(\ell)} (h_a^{(\ell-1)*})^\top + b_a^{(\ell)*} (h_a^{(\ell-1)} - h_a^{(\ell-1)*})^\top \right]. \quad (34)$$

These expressions contain no g . Next compute $V_1 = F_1$ and $V_w = F_w$, and run a directional forward pass,

$$\dot{z}_a^{(1)} = V_1 x_a / \sqrt{d}, \quad \dot{z}_a^{(\ell)} = V_\ell h_a^{(\ell-1)} + \widehat{W}^{(\ell)} \dot{h}_a^{(\ell-1)}, \quad \dot{h}_a^{(\ell)} = \phi'(z_a^{(\ell)}) \odot \dot{z}_a^{(\ell)},$$

which gives $\dot{r}$ and $\dot{\rho}$. A reverse directional pass then differentiates the backward recurrence, using ϕ'' for $\ell \geq 2$, and yields

$$\dot{b}_a^{(\ell)} = \frac{\dot{r}_a \delta_a^{(\ell)} + r_a \dot{\delta}_a^{(\ell)}}{\rho} - b_a^{(\ell)} \frac{\dot{\rho}}{\rho}.$$

Only then do we evaluate $g = \rho + \|\dot{\Psi}\|$ and, finally, (23). Matrix and transpose actions use the same reconstructed operator. In practice one factors G and solves linear systems; an explicit inverse is unnecessary.

The joint clock is more regular in approximation coordinates, but it is not automatically better at low order. At $P = 1$ the polynomial space consists of constants, so changing the coordinate alone leaves the physical constant-mode closure unchanged. Higher orders pay for Gram solves and derivative passes, and the theoretical constants can be large.

C A frozen-span closure of the population flow

This appendix records a closure of the population flow itself, for the smallest case allowed by the contract: two hidden layers, $\phi = \tanh$, inputs on a circle ($d = 2$, $x/\sqrt{d} \in S^1$) with labels $|y| \leq Y$. The population state is the first-layer weight row $W^{(1)} \in L^2(\Omega_1; \mathbb{R}^2)$, the readout $w \in L^2(\Omega_2)$ and the middle operator $W^{(2)} = W_0^{(2)} + (W^{(2)} - W_0^{(2)})$, where $W_0^{(2)}$ is the initialized Gaussian operator and the learned part is Hilbert–Schmidt. The flow is (2) with $\frac{1}{m} \sum_a$ replaced by integration against a training law μ on inputs and labels, and the reverse action is always the true adjoint $(W^{(2)})^*$. We work on $0 \leq t \leq T = 1/200$ and on a Wasserstein neighborhood of the two-point task $\nu_* = \frac{1}{2} \delta_{(\sqrt{2}e_1, +1)} + \frac{1}{2} \delta_{(\sqrt{2}e_2, -1)}$; the neighborhood contains perturbed, nonorthogonal, nonatomic and noisy input laws, and neither its radius nor the interval shrinks with the closure order.

The compressed state. Enumerate the finite expressions built from constants, the Gaussian initial weights, rational affine combinations, $\sin$, $\cos$, ϕ , bounded products, and the initialized actions $W_0^{(2)}$ and $(W_0^{(2)})^*$. At order N this gives, on each population, a finite vector of *observables of the initialization*, which we normalize by a ridge-regularized Gram matrix so that they have unit scale. The saved state is then

- the joint law, on population 1, of the current first-layer weights $W^{(1)}$, their initial values, and the N observables;
- the joint law, on population 2, of the current readout w and the N observables;

- one finite matrix M representing the middle-layer action $W^{(2)}$ between the two observable vectors, initialized from $W_0^{(2)}$.

The closure evolves this state by the population flow itself, with $W^{(2)}$ replaced by the finite operator “project onto population-1 observables, apply M , expand in population-2 observables” and $(W^{(2)})^*$ by its transpose (details below). It is autonomous, uses only the current state and the training law, and is restartable from its own saved state. Initial and current values are stored jointly, so same-neuron quantities such as feature displacement are genuine observables of the state.

Theorem C.1 (Autonomous population closure). *For every separately fixed training law in the stated neighborhood of ν_* and every order N , the closure is uniquely well posed on $[0, T]$ and restartable from its saved state, and as $N \rightarrow \infty$ its prediction f_N converges to the population prediction f uniformly on $[0, T]$ and over the whole input circle. Every separately fixed same-population joint observation—initial and current hidden pairs, feature displacements, forward and adjoint actions, bounded nonlinear pushforwards and quadratic contractions—converges uniformly on $[0, T]$ in Wasserstein distance W_2 .*

The proof (below) controls the distance between closure and population flow through the error of the finite observable basis; unbounded backward fields force a truncation that turns Gronwall’s inequality into an Osgood inequality, whose solution still tends to zero with the basis error.

Proposition C.2 (Both layers learn, nonlinearly). *On a Wasserstein neighborhood of ν_* there is a common time at which both hidden populations have strictly positive mean-square displacement of their activations from initialization, and strictly positive nonaffinity $\inf_{\alpha, \beta} \mathbb{E}|\phi(z) - \alpha z - \beta|^2$ of their preactivations.*

The small-time expansion of the flow at ν_* (below) shows that both hidden displacements are nonzero at order t^2 , a Gaussian preactivation has positive nonaffinity, and continuity extends both properties to a neighborhood. The closure therefore converges on a family where all three properties of the introduction hold.

What this compresses, and why it is not the main result. The learned operator $W^{(2)} - W_0^{(2)}$ of the population flow is replaced by a finite matrix between two populations: an operator ODE becomes a population ODE plus a finite matrix. But the matrix acts between *frozen* spans of initialization observables, so the learned middle layer can only rotate within a span fixed at initialization, while $W^{(1)}$ and w move freely. This is a frozen-dictionary closure of the middle layer, and it shows the characteristic limitations of that class: convergence comes only from the spans becoming dense, there is no rate in N , and the guarantee holds on a short horizon near a reference task. The two laws are continuum objects, so the state is not a finite list of scalars. The frozen-dictionary experiments of Section 7 show the same limitation at finite width, and the response memories of Section 6, whose atoms coevolve with the features, are the answer to it.

Details

Population flow. With $\phi = \tanh$, $d = 2$ and inputs $x \in \sqrt{2}S^1$, the population fields are $z^{(1)}(x) = W^{(1)} \cdot x / \sqrt{d}$, $h^{(1)} = \phi(z^{(1)})$, $z^{(2)}(x) = W^{(2)} h^{(1)}(x)$, $h^{(2)} = \phi(z^{(2)})$, $\delta^{(2)}(x) = w \phi'(z^{(2)}(x))$, $\delta^{(1)}(x) = \phi'(z^{(1)}(x)) (W^{(2)})^* \delta^{(2)}(x)$ and $f(x) = \mathbb{E}_2[w h^{(2)}(x)]$; with $r = f - y$ and a training law μ ,

$$\dot{W}^{(1)} = -2 \int r \delta^{(1)} \frac{x}{\sqrt{d}} d\mu, \quad \dot{W}^{(2)} = -2 \int r \delta^{(2)} \otimes h^{(1)} d\mu, \quad \dot{w} = -2 \int r h^{(2)} d\mu.$$

Observables. Let $\psi_1 \in \mathbb{R}^{N_1}$ and $\psi_2 \in \mathbb{R}^{N_2}$ be the order- N vectors of initialization observables on the two populations, ridge-normalized by $\psi_\ell \mapsto (\mathbb{E}_\ell[\psi_\ell \psi_\ell^\top] + 2^{-N}I)^{-1/2} \psi_\ell$, so that the projections $g \mapsto \psi_\ell^\top \mathbb{E}_\ell[\psi_\ell g]$ are contractions converging strongly to the identity on the observable spaces. The initialized middle action is recorded in $M_0 = \mathbb{E}_2[\psi_2 (W_0^{(2)} \psi_1)^\top]$; the forward and adjoint actions of $W_0^{(2)}$ on observables are drawn from one correlated Gaussian source, which is what makes the two orientations one operator and its adjoint.

Closure. The state is $\text{Law}(\psi_1, W_0^{(1)}, W^{(1)})$ on population 1, $\text{Law}(\psi_2, w)$ on population 2, and $M \in \mathbb{R}^{N_2 \times N_1}$, initialized with $W^{(1)} = W_0^{(1)}$, $w = 0$ and $M = M_0$. The closure is the population flow above with $W^{(2)}$ replaced by

$$W_N^{(2)} g = \psi_2^\top M \mathbb{E}_1[\psi_1 g], \quad (W_N^{(2)})^* g = \psi_1^\top M^\top \mathbb{E}_2[\psi_2 g],$$

$$\dot{M} = -2 \int r_N \mathbb{E}_2[\psi_2 \delta^{(2)}] \mathbb{E}_1[\psi_1 h^{(1)}]^\top d\mu,$$

so that $z_N^{(2)}(x) = \psi_2^\top M \mathbb{E}_1[\psi_1 h^{(1)}(x)]$ and $f_N(x) = \mathbb{E}_2[w h_N^{(2)}(x)]$; the weights $W^{(1)}$ and w move by the flow equations with r_N and $\delta_N^{(1)}$, which transports the two laws. The represented operator is $W_N^{(2)}$, whose learned part is $M - M_0$ written between observable spaces.

Well-posedness. For fixed N the characteristic equations are locally Lipschitz. The loss obeys $\frac{d}{dt} \mathcal{L}_N = -\left\| \dot{W}^{(1)} \right\|_2^2 - \left\| \dot{M} \right\|_F^2 - \|\dot{w}\|_2^2$, and $\mathcal{L}_N(0) \leq Y^2$ gives $\|w(t)\|_\infty \leq 2Yt$, $\|M(t) - M_0\|_F \leq 2Y^2 t^2$, $\left\| W_N^{(2)}(t) \right\|_{\text{op}} \leq 2 + 2Y^2 t^2$ and $\|W^{(1)}(t)\|_2 \leq \sqrt{2} + 4Y^2 t^2 + 2Y^4 t^4$, which exclude finite-time escape. The joint laws contain the correlations between observables and current weights, so the saved state suffices to restart.

Convergence. Place closure and population flow on common probability spaces and set

$$e_N = \left\| W_N^{(1)} - W^{(1)} \right\|_2 + \left\| (W_N^{(2)} - W_{N,0}^{(2)}) - (W^{(2)} - W_0^{(2)}) \right\|_{\text{HS}} + \|w_N - w\|_2,$$

where $W_{N,0}^{(2)}$ is the closure's initial operator. The basis error

$$\varepsilon_N = \sup_{t,x} \left\| (W_{N,0}^{(2)} - W_0^{(2)}) h^{(1)}(t, x) \right\|_2 + \sup_{t,x} \left\| (W_{N,0}^{(2)} - W_0^{(2)})^* \delta^{(2)}(t, x) \right\|_2$$

$$+ \sup_t \left\| \Pi_2 \dot{W}^{(2)}(t) \Pi_1 - \dot{W}^{(2)}(t) \right\|_{\text{HS}},$$

with Π_ℓ the observable projections, tends to zero by strong convergence of the projections, compactness of the population response curves in (t, x) , and density of finite-rank operators. Forward subtraction gives, uniformly in x , $\left\| h_N^{(1)} - h^{(1)} \right\|_2 \leq e_N$ and $\left\| h_N^{(2)} - h^{(2)} \right\|_2 + |f_N - f| \leq C(e_N + \varepsilon_N)$. Backward subtraction multiplies a changed gate by an unbounded reference field; with the exponential tail bound of the reference trajectory, truncating at level R gives

$$e'_N \leq C(1 + R)(e_N + \varepsilon_N) + C e^{-aR}.$$

Choosing $R = 1 + a^{-1} \log(1/v)$ with $v = e_N + \varepsilon_N + \eta$ yields $v' \leq Lv \log(e/v)$, hence

$$e_N(t) + \varepsilon_N + \eta \leq e^{1-\alpha(t)} (\varepsilon_N + \eta)^{\alpha(t)}, \quad \alpha(t) = e^{-Lt}.$$

Letting $\eta \downarrow 0$ and $N \rightarrow \infty$ gives $\sup_{t \leq T} e_N \rightarrow 0$, and the forward bounds give uniform convergence of f_N on $[0, T] \times S^1$. The same induction over observation programs controls every admissible joint observation in L^2 , and the canonical coupling bounds its W_2 distance by the L^2 distance.

Feature learning at ν_* . Write $g_a = W_0^{(1)} \cdot e_a$ for the initial preactivations at the two inputs, $h_a = \phi(g_a)$, $\xi_a = W_0^{(2)} h_a$ and $S = \phi(\xi_1) - \phi(\xi_2)$. With $U_a = S \phi'(\xi_a)$, $P_a = (W_0^{(2)})^* U_a$ and $V_a = \phi'(g_a)^2 P_a$, the population flow satisfies $w(t) = tS + o(t)$, $h^{(1)}(t, e_a) - h_a = \frac{y_a}{2} t^2 V_a + o(t^2)$ and $h^{(2)}(t, e_a) - \phi(\xi_a) = \frac{y_a}{2} t^2 \phi'(\xi_a) (\mathbb{E}[\phi^2(G)] U_a + W_0^{(2)} V_a) + o(t^2)$ for a standard Gaussian G . Adjunction gives $\mathbb{E}[h_a P_a] = y_a \mathbb{E}[\xi_a \phi(\xi_a) \phi'(\xi_a)] \neq 0$, so both hidden displacements are nonzero at order t^2 ; a nondegenerate Gaussian preactivation has positive nonaffinity since $\tanh$ is not affine on any interval; and continuity of the flow in the training law extends both properties to a neighborhood of ν_* .

D Exact compression in a deep linear benchmark

Take one datum $x = 1$, a label y , three linear hidden layers with order-one initialization, mobilities $(n\kappa, \kappa, \kappa, n\kappa)$, and the normalized factors $u = W^{(1)}/\sqrt{n}$, $v = W^{(4)}/\sqrt{n}$, so that $f_n = v^\top W^{(3)} W^{(2)} u$. Packing the four factors into the cyclic block matrix $\mathcal{C}_n$ with blocks $v^\top, u, W^{(2)}, W^{(3)}$ below and above the diagonal gives $f_n = \frac{1}{4} \text{Tr } \mathcal{C}_n^4$, and the feature derivative of $\mathcal{C}_n$ is $(\mathcal{C}_n^\top)^3$.

Theorem D.1 (Exact operator closure). *There are a separable Hilbert space, a bounded initialized source $\mathcal{C}_0$ and a unique global trace-class increment $Q(t)$ with*

$$\mathcal{C} = \mathcal{C}_0 + Q, \quad f = \frac{1}{4} \text{Tr } \mathcal{C}^4, \quad \dot{Q} = -2\kappa(f - y)(\mathcal{C}_0^* + Q^*)^3, \quad Q(0) = 0.$$

The kernel is $K = \text{Tr}[\mathcal{C}^3(\mathcal{C}^)^3]$, the residual obeys $\dot{r} = -2\kappa r K$, and $\|Q(t)\|_1 \leq 2|y|\sqrt{\kappa t}$. For every fixed T , finite gradient flow, and gradient descent for every step sequence tending to zero, converge in probability to $(f, K, \mathcal{L})$ uniformly on $[0, T]$.*

The proof builds $\mathcal{C}_0$ as the limit of initialized word statistics on a colored Fock space, obtains global existence from trace-norm control, passes to the width limit through finite-word Picard iterates, and treats gradient descent as an exact Euler step for Q .

Theorem D.2 (Restricted scalar nonclosure). *In the same model, no encoder formed from a fixed finite list of complete tensor contractions of $(u, W^{(2)}, W^{(3)}, v)$ with connected-graph degree bounded independently of width admits a width-independent polynomial autonomous ODE (or finite-order local polynomial PDE) and a polynomial output readout reproducing the output dynamics on a nonempty open set of states for all sufficiently large widths.*

The $(2j - 1)$ st feature derivative of the output contains the connected contraction

$$u^\top ((W^{(2)})^\top (W^{(3)})^\top W^{(3)} W^{(2)})^j u$$

of degree $4j + 2$. Polynomial operations on a bounded-degree alphabet form only disjoint unions of connected components already present, and nonisomorphic contraction graphs are linearly independent at large width, so this term cannot be synthesized. The theorem does not exclude arbitrary encoders, unrestricted fields, or closures tailored to a single initialized trajectory.